BIO II printed notes: Ch 23

Ruling out answers (Unit 1 pattern):
1. Answers with “only”, “all”, “always”, “exclusively”, “must” or “every” were wrong every time (22 of 22).
2. Drop answers that put a term from another chapter into the wrong organism (e.g. peptidoglycan in a plant).
3. The believable wrong answer is usually the look-alike term from the same slide: check the look-alike table.
4. Questions follow slide order, so the questions before and after tell you which section to look in.
5. Diagram questions ask about the process (n or 2n, where meiosis happens), not label names.

Chapter 23: Plant Evolution and Diversity

23.1 Ancestry and Features of Land Plants
23.1.3 Ancestry and Features of Land Plants 1
23.1.4 Characteristics of Plants 1
23.1.5 The Ancestry of Land Plants 1
23.1.6 Evolutionary History of Plants 1
23.1.7 Embryo Difference in Green Algae and Land Plants 1
23.1.8 Charophytes 2
23.1.9 Adaptation to Land 2
23.1.10 Leaf Adaptation to Prevent Desiccation 2
23.1.11 Land Plants 2
23.1.12 Alternation of Generations 2
23.1.13 Alternation of Generations in Land Plants 2
23.1.14 Dominant Generation 2
23.1.15 Reduction in the Size of the Gametophyte 2

23.2 Evolution of Bryophytes: Colonization of Land
23.2.16 Evolution of Bryophytes: Colonization of Land 2
23.2.17 Liverworts 2
23.2.18 Liverwort, Marchantia 2
23.2.19 Hornworts 2
23.2.20 Hornwort, Anthoceros sp. 2
23.2.21 Mosses 2
23.2.22 Life Cycle of Moss 2
23.2.23–30 Moss Life Cycle 2

23.3 Evolution of Lycophytes: Vascular Tissue
23.3.31 Evolution of Lycophytes: Vascular Tissue 3
23.3.32 The Origin of Vascular Plants 3
23.3.33 A Cooksonia Fossil 3
23.3.34 Lycophytes 3
23.3.35 Lycophytes Example 3

23.4 Evolution of Pteridophytes: Megaphylls
23.4.36 Evolution of Pteridophytes: Megaphylls 3
23.4.37 Microphylls and Megaphylls 3
23.4.38 Horsetails 3
23.4.39 Horsetail, Equisetum 3
23.4.40 Whisk Ferns 3
23.4.41 Whisk Fern, Psilotum 3
23.4.42 Ferns 3
23.4.43 Fern Life Cycle 3
23.4.44 Diversity of Ferns 3
23.4.45–52 Fern Life Cycle 3
23.4.53 The Uses of Ferns 4

23.5 Evolution of Seed Plants: Full Adaptation to Land
23.5.54 Evolution of Seed Plants: Full Adaptation to Land 4
23.5.55 Seed Plants 4
23.5.56 Gymnosperms and Angiosperms 4
23.5.57 Gymnosperms 4
23.5.58 Conifers 4
23.5.59 Conifer Examples 4
23.5.60–68 Pine Life Cycle 4
23.5.69 Uses of Conifers 4
23.5.70 Cycads 4
23.5.71 African Cycad 4
23.5.72 Ginkgoes 4
23.5.73 Ginkgo biloba, a Native of China 4
23.5.74 Gnetophytes 4
23.5.75 Ephedra, a Type of Gnetophyte 5
23.5.76 Welwitschia, a Type of Gnetophyte 5
23.5.77 Angiosperms 5
23.5.78 Origin and Radiation of Angiosperms 5
23.5.79 Evolution of Flowering Plants: Amborella trichopoda 5
23.5.80 Monocots and Eudicots 5
23.5.81 Key Features of Monocots and Eudicots 5
23.5.82 Flower Diversity 5
23.5.83 The Flower 5
23.5.84 Parts of Flower 5
23.5.85 Generalized Flower 5
23.5.86–88 Flowering Plant Life Cycle 5
23.5.89 Life Cycle of Flowering Plant 5
23.5.90 Flowers and Diversification 6
23.5.91 Other Flower Terminology 6

Chapter 23: Plant Evolution and Diversity

23.1 Ancestry and Features of Land Plants

23.1.3

- Plants are multicellular, photosynthetic eukaryotes
- Plant systematists use molecular and morphological information to classify plants
  - Plants are thought to have evolved from a freshwater green algae about 590 MYA
  - Like plants, green algae
    - Contain chlorophylls and accessory pigments
    - Store excess carbohydrates as starch
    - Have cellulose in their cell walls
  - Difference between plants and algae: besides protecting the zygote, plants also protect and nourish the embryo

23.1.4 Characteristics of Plants

- DOMAIN: Eukarya SUPERGROUP: Archaeplastida KINGDOM Plantae
- CHARACTERISTICS Multicellular; apical tissue produces specialized tissues alternation-of-generations life cycle: protection of the embryo: gametangia produces gametes waxy cuticle prevents water loss
- Nonvascular Plants The Bryophytes Liverworts, hornworts, and mosses are low lying and live in moist environments. Dominant gametophyte produces flagellated sperm unbranched, dependent sporophyte produces windblown spores.
- Seedless Vascular Plants - The Lycophytes and Pteridophytes Dominant branched sporophyte has vascular tissue: Lignified xylem transports water, and phloem transports organic nutrients: typically has roots, stems and leaves sporangia borne on sides of leaves that produce windblown spores Independent and separate gametophyte produces flagellated sperm. Lycophyte leaves are microphylls with a single, unbranched vein. Pteridophyte leaves are megaphylls with branched veins.
- Seeded Vascular Plants–The Gymnosperms and Angiosperms Leaves are megaphylls: dominant sporophyte produces heterospores that become dependent male and female gametophytes Male gametophyte is pollen grain, and female gametophyte occurs within ovule, which becomes a seed. Gymnosperms are cone bearing: sporophyte bears pollen cones. which produce windblown pollen (male gametophyte), and seed cones which produce seeds Angiosperm sporophytes bear flowers, which produce pollen grain and bear ovules within an ovary Following double fertilization, ovules become seeds that enclose a sporophyte embryo and endosperm. Fruit develops from ovary

23.1.5 The Ancestry of Land Plants

- Land plants are most closely related to freshwater green algae known as charophytes
  - Charophytes and land plants are in the same clade
    - Ancestors of land plants were most closely related to the charophytes Chara and Coleochaete
      - Most important to the evolution of plants, charophytes protect the zygote

23.1.6 Evolutionary History of Plants

Fig: Common green algal ancestor leads to vascular (seed) through flowering plants and gymnosperms. Common green algal ancestor leads to vascular (seedless) through ferns and allies and lycophytes. Common green algal ancestor leads to nonvascular (bryophytes) through hornworts, mosses and liverworts. Common green algal ancestor leads to charophytes.

23.1.7 Embryo Difference in Green Algae and Land Plants

Fig: A) Microscopic view of embryo of Oedogonium is depicted a tube-like alga with unprotected embryo. B) Microscopic view shows a nourished and protected embryo of Capsella encapsulated in seed.

23.1.8 Charophytes

Fig: A) Chara is depicted as stem-like and leaf-like structures. B) Micrograph shows concentric rings of green, uninucleated cells.

Textbook caption: The charophytes (represented here by Chara and Coleochaete) are the green algae most closely related to the land plants.

23.1.9 Adaptation to Land

- Being on land required plants to evolve mechanisms to minimize dehydration, move water and nutrients against gravity, and damage from UV radiation:
  - Waxy covering called a cuticle protects most land plants from desiccation
    - Cuticle covers exposed surfaces and is relatively impermeable
    - Small openings, called stomata, are found mostly on the underside of leaves and allow for gas exchange between the plant and the environment
  - To move water against the force of water many plants have tracheids, specialized cells with proteins that resist gravity and facilitate the upward transport of water and minerals

23.1.10 Leaf Adaptation to Prevent Desiccation

Fig: A) Leaves have a waterproof cuticle: Enlarged diagram shows the presence of white cuticle. B) Falsely colored scanning electron micrograph of leaf surface: Micrograph (enlarged 214 times) shows presence of eye shaped stomata across the surface of leaf.

23.1.11 Land Plants

- Land plants have tissues which reduces water loss to the environment because they have a lower surface-area-to-volume ratio than do branched filaments
- To reduce the effects of UV radiation, plants have both a haploid and diploid generation
  - Diploid generation helps to hide the effect of a single deleterious allele
  - Evolutionary shift toward a more dominant diploid generation also allows for greater genetic variability in land plants

23.1.12 Alternation of Generations

- All land plants exhibit alternation of generations:
  - Plant alternates between a multicellular haploid (n) phase and a multicellular diploid (2n) phase
    - Multicellular haploid individuals are called gametophytes
    - Gametophytes produce gametes (eggs and sperm) by mitosis within tissue called gametangia
      - Male gametangia are called antheridia
      - Female gametangia are called archegonia
      - Eggs and sperm combine to form diploid sporophytes
  - Multicellular diploid individuals are called sporophytes
    - Sporophytes produce spores by meiosis within a structure called the sporangium
    - Spores are released and if they land on suitable substrate, germinate and grow by mitosis into gametophytes

23.1.13 Alternation of Generations in Land Plants

Fig: A sporophyte (2N) gives rise to spores through meiosis and a gametophyte (1N) gives rise to gametes that fuse in fertilization to form a zygote. In diploid phase, zygote(2n) on mitosis leads to sporophyte (2n), which further leads to sporangium (2). In haploid phase (n), meiosis leads to spores (n), which on mitosis leads to gametophyte (n). Further, on mitosis, gametes (n) are formed which on fertilization leads to formation of zygote in diploid phase.

23.1.14 Dominant Generation

- Land plants differ as to which generation is dominant (conspicuous)
  - In early land plants like moss, the gametophyte is dominant
  - In all other plants such as ferns, pine trees, and flowering plants, the sporophyte is dominant
    - Shift to sporophyte dominance is adaptation to land
      - Only sporophytes have vascular systems
      - As the sporophyte becomes dominant, the gametophyte becomes reduced in size and dependent on sporophyte

23.1.15 Reduction in the Size of the Gametophyte

Fig: Illustration shows reduction in the size of the gametophyte and increase in sporophytes from moss to angiosperm. The data is as follows: Gametophyte (n): Moss: rhizoids. Fern: rhizoids Gymnosperm: blank Angiosperm: blank Sporophytes (2n): Moss: spores Fern: spores and roots Gymnosperm: seed and roots Angiosperm: seed and roots.

Textbook caption: Notice the reduction in the size of the gametophyte and the increase in the size of the sporophyte among these representatives of today's land plants. This trend occurred as these plants became adapted for life on land. In the moss and fern, spores disperse the gametophyte. In gymnosperms and angiosperms, seeds disperse the sporophyte.

23.2 Evolution of Bryophytes: Colonization of Land

23.2.16

- Nonvascular plants
  - First plants to colonize land
    - Absence of vascular tissue results in low-lying plants that
      - Lack specialized means of transporting water and organic nutrients
      - Do not have true roots, stems, and leaves
  - Gametophyte dominant generation
    - Produces eggs in archegonia
    - Produces flagellated sperm in antheridia
    - Sperm swim to egg in film of water

23.2.17 Liverworts

- Liverwort gametophytes (n) are divided into two groups:
  - Thallose liverworts: have a flattened body known as a thallus
  - Leafy liverworts: superficially resemble mosses
- Some liverworts like Marchantia contain rhizoids
  - Hairy extensions that anchor plant in soil and aid in absorption
- Liverworts can reproduce either sexually or asexually
  - Gemmae are groups of cells that detach from the thallus and start a new plant (asexual)
  - Sexual reproduction depends on two types of stalks bearing either antheridia (n) or archegonia (n)
    - Sporophyte (2n) grows from a fertilized egg within the archegonia
    - Sporophyte grows a capsule which produces and releases spores (n)

23.2.18 Liverwort, Marchantia

Fig: a) Microscopic view shows thallus present between gemma cup and rhizoids. b) Male gametophytes bearing antheridia is depicted as sac-like structure of sterile cells. c) Female gametophyte bearing archegonia is depicted as umbrella shaped structure on top of a stalk.

23.2.19 Hornworts

- Have small sporophytes rising out of thin gametophytes that carry on photosynthesis
- Sporophyte has one chloroplast per cell
- Most live in moist, well-shaded areas.
- Symbiotic relationship with cyanobacteria allow nitrogen fixation from the air
- Can bypass alternation of generations by reproducing asexually through fragmentation

23.2.20 Hornwort, Anthoceros sp.

Fig: The “horns” of a hornwort are sporophytes that grow continuously from a base anchored in gametophyte tissue.

23.2.21 Mosses

- Largest phyla of nonvascular plants
- Mostly found in moist environments
- In dry environments they may become shriveled, turn brown, and look dead but when it rains they resume metabolic activity and turn green
- Three distinct groupings of mosses:
  - Peat mosses, granite mosses, true mosses
    - Peat moss has great commercial and ecological importance to humans
      - Used to improve the water-capacity of soil
      - Was used as wound dressing on battlefields

23.2.22 Life Cycle of Moss

- Gametophyte (n) of a moss is called the protonema
  - Precedes and produces upright, leafy shoots that sprout rhizoids
  - Shoots bear either antheridia (sperm-producing) or archegonia (egg-producing)
  - Sporophyte is enclosed in female gametophyte tissue
  - Sporophyte has three parts:
    - A foot, a stalk, and an upper capsule
      - Capsule contains the sporangium, where spores are produced
      - Moss sporophyte is always attached to the gametophyte

Textbook: The protonema is only the first, thread-like stage of the moss gametophyte (a spore germinates into a male or female protonema, 23.2.30); it then produces the leafy shoots.

23.2.23 Moss Life Cycle 1

Fig: Step 1 (mature gametophytes) of the moss life cycle; full description at 23.2.30.

23.2.24 Moss Life Cycle 2

Fig: Step 2 (fertilization) of the moss life cycle; full description at 23.2.30.

23.2.25 Moss Life Cycle 3

Fig: Step 3 (zygote) of the moss life cycle; full description at 23.2.30.

23.2.26 Moss Life Cycle 4

Fig: Step 4 (sporophyte) of the moss life cycle; full description at 23.2.30.

23.2.27 Moss Life Cycle 5

Fig: Step 5 (spores) of the moss life cycle; full description at 23.2.30.

23.2.28 Moss Life Cycle 6

Fig: Step 6 (spore dispersal) of the moss life cycle; full description at 23.2.30.

23.2.29 Moss Life Cycle 7

Fig: Step 7 (immature gametophyte) of the moss life cycle; full description at 23.2.30.

23.2.30 Moss Life Cycle 8

Fig: The details of the moss life cycle between fertilization and meiosis are as follows: 1. The mature gametophytes: In mosses, the leafy gametophyte shoots bear either antheridia or archegonia, where gametes are produced by mitosis. 2. Fertilization: Flagellate sperm produced in antheridia swim in external water to archegonia, each bearing a single egg. 3. The zygote: The zygote and developing sporophyte are retained within the archegonium. 4. The sporophyte: The mature sporophyte has a foot buried in female gametophyte tissue, a stalk, and an upper capsule (the sporangium), where meiosis occurs and spores are produced. 5. The spores: When the lid (operculum) of a capsule falls of, the spores are mature. One or two rings of teeth project inward from the margin of the capsule. The teeth close the opening, except when the weather is dry. 6. Spore dispersal: Spores are released when they are most likely to be dispersed by air currents. 7. The immature gametophyte: A spore germinates into a male or female protonema, the first stage of the male and the female gametophytes.

23.3 Evolution of Lycophytes: Vascular Tissue

23.3.31

- Vascular plants (tracheophytes) dominate the natural terrestrial landscape
  - Can achieve great heights because of roots and vascular tissue and nutrient-conducting tissue:
    - Xylem conducts water and dissolved minerals up from roots
      - Lignin strengthens cell walls of conducting cells in xylem
    - Phloem conducts sucrose and other organic compounds throughout the plant

23.3.32 The Origin of Vascular Plants

- Fossil record indicates that the first vascular plants, such as Cooksonia, were more likely bushes than trees
  - Branching of Cooksonia was significant for two reasons:
    - More sporangia were produced, and therefore many more spores
    - Indicative of vascular tissue
  - Sporangia of Cooksonia produced windblown spores
    - This classifies it (and other lycophytes and pteridophytes) as seedless vascular plants
  - Lycophytes and all other vascular plants have a dominant sporophyte generation

23.3.33 A Cooksonia Fossil

Fig: The fossil, which is no more than a few centimeters tall and terminated in sporangia is evidence of vascular plant evolution.

23.3.34 Lycophytes

- Roots evolved as lower extensions of the stem
- Leaves are microphylls
  - Have only one strand of vascular tissue
  - Most likely evolved as simple side extensions of the stem
- Cone-shaped structure called a strobili produce spores by meiosis
- Some lycophytes have spores that grow into one type of gamete (homosporous), others produce two types of spores (heterosporous)
- Microspores (become male gametophyte) and megaspores (become female gametophyte)
- Three main groups of lycophytes:
  - Ground pines (Lycopodium), spike mosses (Selaginella), and quillworts (Isoetes)

Textbook: Homosporous means making one type of spore, which grows into a gametophyte that makes both eggs and sperm. The slide's "one type of gamete" should read "one type of spore."

23.3.35 Lycophytes Example

Fig: A) A spike moss, Selaginella. B) The upper part of club moss consists of strobili (which further consists of sporangia units) and leaves (microphylls), the middle parts consists of branches and the lower part consists of aerial stem, lateral rhizome and root.

23.4 Evolution of Pteridophytes: Megaphylls

23.4.36

- Pteridophytes are seedless vascular plants
  - Dominant from late Devonian through Carboniferous period
  - Include ferns and their allies (horsetails and whisk ferns)
  - Have megaphylls (broad leaves with several strands of vascular tissue)
    - Allow plants to efficiently collect solar energy, allowing the production of more food and possibly more offspring
      - Resulted in increased fitness for plants

23.4.37 Microphylls and Megaphylls

Fig: A) Microphyll: It consists of a single strand of vascular tissue. Megaphyll: It consists of branched vascular tissue. B) Megaphyll evolution process: The branched stem system leads to one branch began to dominate the stem system, leading to the side branches flattened into a single plane which further leads to tissue filled in the spaces between the side branches and megaphyll leaf.

23.4.38 Horsetails

- Horsetails consist of one genus, Equisetum, and approximately 15 species of seedless vascular plants
  - Most horsetails inhabit wet, marshy environments
  - Equisetum can be branched or unbranched, with leaves that may have been megaphylls at one time but now are reduced and form whorls at the nodes
  - Branches bear resemblance to a horse's tail
  - Many have strobili at the tips of stems
  - Spores germinate into inconspicuous and independent gametophytes
  - Equisetum spores are sensitive to humidity and are "spring-loaded"
    - Spores have been known to be ejected and travel upward of 2 mph
  - Stems are tough and rigid because of silica deposited in cell walls

23.4.39 Horsetail, Equisetum

Fig: Diagram of horsetail plant shows labels for root, rhizome, leaves, node, branches, and strobilus.

Textbook caption: Whorls of branches and tiny leaves are at the nodes of the stem. Spore-producing sporangia are borne in strobili.

23.4.40 Whisk Ferns

- Whisk ferns consist of two genera: Psilotum and Tmesipteris
  - Resemble whisk brooms because they have no leaves
  - Horizontal rhizome gives rise to aboveground stems that repeatedly fork
  - Pumpkin-shaped sporangia are on short side branches
  - Tmesipteris species have appendages resembling reduced megaphylls

23.4.41 Whisk Fern, Psilotum

Fig: Close-up of sporangium (left). Diagram of whisk fernroot plant (right) shows labels for rhizome, root, aerial stem, scale and sporangium.

Textbook caption: Psilotum has no leaves; the branches carry on photosynthesis. The sporangia are yellow.

23.4.42 Ferns

- Approximately 11,000 species
- Most abundant in warm, moist, tropical regions
- Range in size from 1 cm in diameter to 20 m in height
- Large, conspicuous megaphylls are called fronds
  - Divided into leaflets
  - Sporangia are in clusters called sori on the undersides of fronds
- Dominant sporophyte produces windblown spores by meiosis within sporangia

23.4.43 Fern Life Cycle

- Windblown spores eventually germinate into gametophytes which produce sperm
- Sperm swim from antheridium to archegonium where fertilization occurs
- When the gametophyte disappears the sporophyte is independent
  - Both generations of a fern are considered to be independent of one another

23.4.44 Diversity of Ferns

Fig: Photos of a tree fern, a feather leaf fern, and a hart's tongue fern.

23.4.45 Fern Life Cycle 1

Fig: Overview (the dominant sporophyte) of the fern life cycle; full description at 23.4.52.

23.4.46 Fern Life Cycle 2

Fig: Step 1 (sporophyte) of the fern life cycle; full description at 23.4.52.

23.4.47 Fern Life Cycle 3

Fig: Step 2 (sporangia) of the fern life cycle; full description at 23.4.52.

23.4.48 Fern Life Cycle 4

Fig: Step 3 (spores) of the fern life cycle; full description at 23.4.52.

23.4.49 Fern Life Cycle 5

Fig: Step 4 (gametophyte) of the fern life cycle; full description at 23.4.52.

23.4.50 Fern Life Cycle 6

Fig: Step 5 (fertilization) of the fern life cycle; full description at 23.4.52.

23.4.51 Fern Life Cycle 7

Fig: Step 6 (zygote) of the fern life cycle; full description at 23.4.52.

23.4.52 Fern Life Cycle 8

Fig: Details of the fern life cycle are shown. 1. The sporophyte: The sporophyte portion of the life cycle is dominant in ferns. 2. The sporangia: In this fern, the sporangia are located within sori, on the underside of the leaflets, and protected by the indusium. 3. The spores: Within a sporangium, meiosis occurs and spores are produced. When a sporangium opens, the spores are released. 4. The gametophyte: A spore germinates into a prothallus (the gametophyte), which typically bears archegonia at the notch and antheridia at the tip between the rhizoids. 5. Fertilization: Fertilization takes place when moisture is present, because the flagellated sperm must swim in a film of water from the antheridia to the egg within the archegonium. 6. The zygote: The resulting sporophyte zygote begins its development inside an archegonium. As the distinctive first leaf appears above the prothallus, and the roots develop below it, the sporophyte becomes visible. 7. The fronds: The sporophyte develops a root-bearing rhizome from which the aerial fronds project.

23.4.53 The Uses of Ferns
- Ferns have economic and medicinal value
  - Edible ferns are used as a food source
  - Native Americans use them as an astringent during childbirth to stop bleeding
  - Maidenhair fern is the source of a cold medicine
  - Ferns are frequently used by florists in decorative bouquets and as ornamental plants in the home and garden
  - Wood from tropical tree ferns is often used as a building material, because it resists decay, particularly by termites

23.5 Evolution of Seed Plants: Full Adaptation to Land

23.5.54
- Seed plants are vascular plants that use seeds during the dispersal phase of their life cycle
  - Seeds contain a sporophyte embryo and stored food within a protective seed coat
    - Seed coat and stored food allow an embryo to survive harsh conditions during long periods of dormancy (arrested state) until environmental conditions become favorable for growth
    - When a seed germinates, the stored food is a source of nutrients for the growing seedling
    - Survival of seeds largely accounts for the dominance of seed plants today

23.5.55 Seed Plants
- Seed plants are heterosporous (have microspores and megaspores)
- Unlike other plant groups, seed plants retain the spores rather than release them into the environment
  - Microspores become male gametophytes (pollen grains)
    - Pollination occurs when wind or a pollinator carries pollen to the female gametophyte
      - Sperm move toward the female gametophyte through a pollen tube
      - Megaspore develops into a female gametophyte within an ovule, which becomes a seed following fertilization
      - No external water is needed to accomplish fertilization

23.5.56 Gymnosperms and Angiosperms
- In gymnosperms (mostly cone-bearing seed plants), the ovules are not completely enclosed by sporophyte tissue at the time of pollination
- In angiosperms (flowering plants), the ovules are completely enclosed within diploid sporophyte tissue (ovary), which becomes a fruit.

23.5.57 Gymnosperms
- Seeds are not enclosed by fruit—"naked" seeds.
- 1,000 species divided into four extant groups:
  - Conifers – Most plentiful
  - Cycads
  - Ginkgoes
  - Gnetophytes

23.5.58 Conifers
- About 630 species
  - Produce cones
  - Tough, needlelike leaves of pines conserve water with a thick cuticle and recessed stomata
  - Sporophyte is dominant
  - Pollen grains are windblown
  - Seed is the dispersal stage
  - Monoecious – A single plant produces both pollen (male reproductive structure) and seed cones (female reproductive structure)

23.5.59 Conifer Examples
Fig: Examples of conifer include pine having pollen and seed cones (a), spruce (needless leaves), and Juniper (tall shrub, scale-like leaves.
Textbook caption: Pine trees are the most common of the conifers. The pollen cones (male) are smaller than the seed cones (female) and produce pollen.

23.5.60 Pine Life Cycle 1
Fig: Overview (the tree) of the pine life cycle; full description at 23.5.68.
Fig: The tree is the diploid (2n) sporophyte, the most prominent structure in the pine life cycle; it bears both pollen cones and seed cones.

23.5.61 Pine Life Cycle 2
Fig: Step 1 (pollen and seed cones) of the pine life cycle; full description at 23.5.68.

23.5.62 Pine Life Cycle 3
Fig: Step 2 (pollen sacs and ovules) of the pine life cycle; full description at 23.5.68.

23.5.63 Pine Life Cycle 4
Fig: Step 3 (microspores and megaspores) of the pine life cycle; full description at 23.5.68.

23.5.64 Pine Life Cycle 5
Fig: Step 4 (pollen grains) of the pine life cycle; full description at 23.5.68.

23.5.65 Pine Life Cycle 6
Fig: Step 5 (mature female gametophyte) of the pine life cycle; full description at 23.5.68.

23.5.66 Pine Life Cycle 7
Fig: Step 6 (zygote) of the pine life cycle; full description at 23.5.68.

23.5.67 Pine Life Cycle 8
Fig: Step 7 (sporophyte and seeds) of the pine life cycle; full description at 23.5.68.

23.5.68 Pine Life Cycle 9
Fig: Details of a typical conifer life cycle are shown. 1. The pollen cones: Typically, the pollen cones are quite small and develop near the tips of lower branches. The seed cones: The seed cones are larger than the pollen cones and are located near the tips of higher branches. 2. The pollen sac: A pollen cone has two pollen sacs (microsporangia) that lie on the underside of each scale. The ovules: The seed cone has two ovules (megasporangia) that lie on the upper surface of each scale. 3. The microspores: Within the pollen sacs, meiosis produces four microspores. The megaspore: Within an ovule, meiosis produces four megaspores; only one survives. 4. The pollen grains: Each microspore becomes a pollen grain, which has two wings and is carried by the wind to the seed cone during pollination. 5. The mature female gametophyte: Only one of the megaspores undergoes mitosis and develops into a mature female gametophyte, having two to six archegonia. Each archegonium contains a single large egg lying near the ovule. 6. The zygote: Once a pollen grain reaches a seed cone, it becomes a mature male gametophyte. A pollen tube digests its way slowly toward a female gametophyte and discharges two nonflagellated sperm. One of these fertilizes an egg in an archegonium, and a zygote results. 7. The sporophyte: After fertilization, the ovule matures and becomes the seed composed of the embryo, reserve food, and a seed coat. Finally, in the fall of the second season, the seed cone, by now woody and hard, opens to release winged seeds. When a seed germinates, the sporophyte embryo develops into a new pine tree, and the cycle is complete.

23.5.69 Uses of Conifers
- Pines are known for their beauty and pleasant smell
  - Make attractive additions to parks and gardens
  - Have medicinal value
    - Needles and bark are rich in vitamins A and C
      - Can be used to create teas that ease symptoms of a cold or cough
  - Large pine seeds, called pine nuts, are sometimes harvested for use in cooking
  - Pine oil is used to scent a number of household and personal care products, such as room sprays and deodorant

23.5.70 Cycads
- Include 10 genera and 300 species of distinctive gymnosperms
- Large, finely divided leaves that grow in clusters at the top of the stem
- Cones can be more than 1 m long
- Pollen and seed cones are on separate plants
- Pollinated by insects rather than wind
  - Pollen tube bursts near egg and multiflagellated sperm swim to reach an egg
- Plentiful in Mesozoic era at the time of the dinosaurs
  - Dinosaurs fed on cycad seeds

23.5.71 African Cycad
- An African cycad

Fig: A cycad exhibits trunk, leaves, with an inset on top-right depicting presence of cones on separate plants.
Textbook caption: Cycads may resemble ferns or palms, but they are cone-producing gymnosperms.

23.5.72 Ginkgoes
- One surviving species is Gingko biloba
  - Dioecious
    - Some trees produce seeds
    - Other trees produce pollen
  - Trees are resistant to pollution and do well along city streets and in city parks
  - Native to China, and in Asia, ginkgo seeds are considered a delicacy
  - Extracts have been used to improve blood circulation
  - Like cycads, the pollen tube of ginkgo bursts to release multiflagellated sperm that swim to the egg

23.5.73 Ginkgo biloba, a Native of China
Fig: A ginkgo tree has broad leaves and fleshy seeds borne at the end of stalk like megasporophylls.

23.5.74 Gnetophytes
- Three living genera with 65 diverse species
- They have similarly structured xylem
- None have archegonia
- Strobili have similar construction
  - Gnetum consists of trees or climbing vines with broad, leathery leaves arranged in pairs
  - Ephedrine, a medicine with serious side effects, is extracted from Ephedra

– Welwitschia, found in southwestern Africa, has only two enormous, straplike leaves that fray as it gets older

23.5.75 Ephedra, a Type of Gnetophyte

Fig: This example is a branched shrub with small, scale-like leaves. Pollen is produced in microsporangia.

23.5.76 Welwitschia, a Type of Gnetophyte

Fig: This example produces 2 straplike leaves that fray and split.

23.5.77 Angiosperms

- Flowering plants
- Called angiosperms because their ovules are always enclosed within diploid tissues
- Exceptionally large and successful group of plants
  - 352,000 known species
- Range in size from tiny duckweed to Eucalyptus, over 100 m tall
- All fruits, vegetables, nuts, herbs, and grains, which are the staples of the human diet, are angiosperms
- Provide us with clothing, food, medicines, and other commercially valuable products

23.5.78 Origin and Radiation of Angiosperms

- Until recently, the earliest known angiosperm fossil was Archaefructus liaoningensis
  - Dates back 135 million years to the Jurassic
  - Archaefructus fossil has fruit and flowers
- Pollen can be used to date the appearance of a seed plant in an area
  - Paleobotanists working in Switzerland discovered 240-million-year-old fossilized pollen from the Triassic
- Living angiosperm of today that might be most closely related to the first angiosperms is Amborella
  - Lives only on the island of New Caledonia in the South Pacific
  - Botanists hope that much may be learned about the early adaptive radiation of angiosperms during the Tertiary period

23.5.79 Evolution of Flowering Plants: Amborella trichopoda

- Amborella trichopoda

Fig: Close-up of a plant shows small, creamy white flowers arranged in inflorescence borne in the axils of foliage leaves.

Textbook caption: Analysis of molecular data suggests that Amborella trichopoda is the most closely related to the first flowering plants.

23.5.80 Monocots and Eudicots

- Most flowering plants belong to one of two classes:
  - Monocotyledones (monocots)
    - Examples include corn, tulips, pineapples, and sugarcane
  - Eudicotyledones (eudicots)
    - Examples include cacti, strawberries, dandelions, poplars, and beans
- Cotyledons are the seed leaves containing nutrients that nourish the embryo
  - Monocots: One cotyledon in seed
  - Eudicots: Two cotyledons in seed

23.5.81 Key Features of Monocots and Eudicots

Monocots	Eudicots
One cotyledon	Two cotyledons
Flower parts In threes or multiples of three	Flower parts In fours or fives or multiples of four or five
Pollen grain with one pore	Pollen grain with three pores
Usually herbaceous	Woody or herbaceous
Usually parallel venation	Usually net venation
Scattered bundles In stem	Vascular bundles in a ring
Fibrous root system	Taproot system

23.5.82 Flower Diversity

Fig: A) Three white colored petals (monocots). B) Five petals flower in white with stripes of pink (eudicot). Regardless of size and shape, flowers, such as this monocot, and eudicot, share certain features.

23.5.83 The Flower

- Although flowers vary widely in appearance, most have certain structures in common:
  - Sepals are collectively called the calyx
    - Protect the flower bud before it opens
    - Sepals may drop off or may be colored like the petals
    - Usually, however, sepals are green and remain attached to the flower stalk.
  - Petals are collectively called the corolla
    - Very diverse in size, shape, and color
    - Petals are often used to attract a particular pollinator

23.5.84 Parts of Flower

- Stamens are the male structures and consist of two parts:
  - Filament: a slender stalk that holds up the anther
  - Anther: saclike container that produces pollen
- Carpels are the female structure and are comprised of three major regions:
  - Stigma, an enlarged, sticky knob
  - Style, a slender stalk
  - Ovary, an enlarged base that encloses one or more ovules
    - Ovule becomes the seed, and the ovary becomes the fruit. Fruit is often instrumental in the distribution of seeds.

23.5.85 Generalized Flower

Fig: A flower has 4 main parts: sepals, petals, stamens (anther and filament), and carpels (stigma, style, ovule, pollen tube, and ovary).

Textbook caption: A flower has four main parts: sepals, petals, stamens, and carpels. A stamen has an anther and filament. A carpel has a stigma, style, and ovary. An ovary contains ovules.

23.5.86 Flowering Plant Life Cycle 1

- Flowering plants are heterosporous, producing two types of spores:
  - Megaspore
    - Located in an ovule within an ovary of a carpel
    - Develops into an egg-bearing female gametophyte, called the embryo sac
      - Usually has seven cells, two have special functions:
        - One is an egg
        - One contains two polar nuclei
          - "Polar" because they come from opposite ends of the embryo sac

23.5.87 Flowering Plant Life Cycle 2

  - Microspores
    - Produced within anthers
    - Become pollen grains that, when mature, are male gametophytes with sperm
    - Mature male gametophyte consists of only three cells: the tube cell and two sperm cells
- During pollination, pollen is transported from the anther to the stigma of a carpel
  - Tube cell produces a pollen tube
  - Two sperm travel through the tube to the micropyle (small opening) of an ovule

23.5.88 Flowering Plant Life Cycle 3

- Flowering plants undergo double fertilization
  - One sperm unites with an egg
    - Forms a diploid zygote
  - Other sperm unites with polar nuclei
    - Forms a triploid endosperm nucleus
- Ovule becomes a seed
  - Contains the embryo (the sporophyte of the next generation)
  - Stores food (endosperm) enclosed within a seed coat
- Fruit is derived from an ovary
  - Some fruits, such as apples provide a moist, fleshy covering
  - Other fruits, such acorns, provide a dry covering for seeds

23.5.89 Life Cycle of Flowering Plant

Fig: Details of a typical flowering plant cycle are shown.1. The stamen: An anther at the top of a stamen has four pollen sacs. Pollen grains are produced in pollen sacs. The carpel: The ovary at the base of a carpel contains one or more ovules. The contents of an ovule change during the flowering plant lifecycle. 2. The pollen sacs: In pollen sacs (microsporangia) of the anther, meiosis produces microspores. The ovules: In an ovule (megasporangium) within an ovary, meiosis produces four megaspores. 3. The microspores: Each microspore in a pollen sac undergoes mitosis to become an immature pollen grain with two cells: the tube cell and the generative cell. The pollen sacs open, and the pollen grains are windblown or transported by an animal carrier, usually to other flowers. This is known as pollination. The megaspores: Inside the ovules of an ovary, three megaspores disintegrate, and only the remaining one undergoes mitosis to become a female gametophyte. 4. The mature male gametophyte: A pollen grain that lands on the carpel of the same type of plant germinates and produces a pollen tube, which grows within the style until it reaches an ovule in the ovary. Inside the pollen tube, the generative cell nucleus divides and produces two nonflagellated sperm. A fully germinated pollen grain is the mature male gametophyte. The mature female gametophyte: The ovule now contains the mature female gametophyte (embryo sac), which typically consists of eight haploid nuclei embedded in a mass of cytoplasm. The cytoplasm differentiates into cells, one of which is an egg and another that contains two polar nuclei. 5. Double fertilization: On reaching the ovule, the pollen tube discharges the sperm. One of the two sperm migrates to and fertilizes the egg, forming a zygote; the other unites with the two polar nuclei, producing a 3n (triploid) endosperm nucleus. The endosperm nucleus divides to form endosperm, food for the developing embryo. 6. The seed: The ovule now develops into the seed, which contains an embryo and food enclosed by a protective seed coat. The wall of the ovary and sometimes adjacent parts develop into a fruit that surrounds the seed(s). 7. The sporophyte: The embryo within a seed is the immature sporophyte. When a seed germinates, growth and differentiation produce the mature sporophyte of a flowering plant.

23.5.90 Flowers and Diversification

- Various ways pollen and seeds can be dispersed have resulted in many different types of flowers:
  - Wind-pollinated flowers are usually not showy
  - Bird-pollinated and insect-pollinated flowers are often colorful
  - Night-blooming flowers attract nocturnal mammals or insects
    - Usually white or cream-colored and aromatic
  - Many flowers are adapted to a specific pollinator
  - Fruits of flowers protect and aid in seed dispersal
    - Utilize wind, gravity, water, and animals for dispersal

23.5.91 Other Flower Terminology

Term	Description of Flower
Complete	All four parts (sepals, petals, stamens, and carpels) present
Incomplete	Lacks one or more of the four parts
Perfect	Has both stamens and (a) carpel(s)
Imperfect	Has stamens or (a) carpel(s), but not both
Inflorescence	A cluster of flowers
Composite	Appears to be a single flower but consists of a group of tiny flowers

Comparison tables

Plant groups 23.1.4, 23.2.16, 23.3.32, 23.3.34, 23.4.36, 23.5.55–57, 23.5.77

Group	Examples	Vascular tissue	Dominant generation	Leaves	Spores	Seeds / fruit	Sperm reach the egg by
Bryophytes (nonvascular)	liverworts, hornworts, mosses	none, so low-lying; no true roots, stems or leaves	gametophyte; sporophyte unbranched, dependent	no true leaves	windblown	none	flagellated sperm swim in a film of water
Lycophytes (seedless vascular)	ground pines, spike mosses, quillworts	yes (lignified xylem, phloem); roots	sporophyte; gametophyte independent, separate	microphylls (one unbranched vein)	windblown, from strobili; homosporous or heterosporous	none	flagellated sperm (need water)
Pteridophytes (seedless vascular)	ferns, horsetails, whisk ferns	yes	sporophyte; fern gametophyte independent	megaphylls (branched veins); whisk ferns have none	windblown; ferns: in sori; horsetails: in strobili	none	flagellated sperm swim in a film of water
Gymnosperms (seed plants)	conifers, cycads, ginkgoes, gnetophytes	yes	sporophyte; gametophytes dependent	megaphylls (pine needles)	heterosporous; spores kept, not released	"naked" seeds, not enclosed by fruit; mostly on cones	pollen tube; no external water (cycads, ginkgo: multiflagellated sperm swim from the burst tube)
Angiosperms (seed plants)	flowering plants: monocots, eudicots	yes	sporophyte; gametophytes dependent	megaphylls	heterosporous; spores kept	seeds enclosed in the ovary, which becomes the fruit	pollen tube carries 2 sperm; double fertilization; no external water

Life cycles compared 23.2.30, 23.4.52, 23.5.68, 23.5.86–89

	Moss	Fern	Pine	Flowering plant
Dominant	gametophyte (the leafy shoots)	sporophyte; both generations independent	sporophyte (the tree, 2n)	sporophyte
Meiosis in	capsule (the sporangium) on top of the sporophyte	sporangia in sori under the leaflets (covered by the indusium)	pollen sacs under pollen-cone scales → 4 microspores; ovules on seed-cone scales → 4 megaspores, 1 survives	anther's pollen sacs → microspores; ovule → 4 megaspores, 3 disintegrate
Gametophyte	spore → protonema (first stage) → leafy shoots with antheridia or archegonia	spore → prothallus: archegonia at the notch, antheridia at the tip	male: winged pollen grain; female: in the ovule, 2–6 archegonia with one egg each	male: pollen grain (tube cell + generative cell → 2 sperm); female: embryo sac (8 nuclei: egg, 2 polar nuclei)
Fertilization	flagellate sperm swim in external water to the archegonia	needs moisture: flagellated sperm swim in a film of water	pollen tube digests toward the egg, releases 2 nonflagellated sperm; one fertilizes the egg	pollen tube grows down the style to the ovule; double fertilization: sperm + egg → 2n zygote, sperm + 2 polar nuclei → 3n endosperm
Then	zygote stays in the archegonium; sporophyte = foot + stalk + capsule, always attached	zygote grows in the archegonium; first leaf and roots appear; rhizome with fronds	ovule → seed (embryo, reserve food, seed coat); in the fall of the second season the woody cone opens	ovule → seed (embryo, endosperm, seed coat); ovary wall → fruit
Dispersed as	spores, by air currents	windblown spores	winged seeds	seeds in fruits (wind, gravity, water, animals)

Land adaptations 23.1.9, 23.1.11, 23.1.14

Problem on land	Solution
drying out	waxy cuticle; tissues (lower surface-area-to-volume ratio than branched filaments)
gas exchange through the cuticle	stomata, mostly on the underside of leaves
moving water up against gravity	tracheids (vascular tissue)
UV damage	a diploid generation, which hides the effect of a single harmful allele
(overall trend)	dominant sporophyte: only sporophytes have vascular systems; the gametophyte shrinks and becomes dependent

Bryophytes

Group	Key features
Liverworts	thallose (flat thallus) or leafy (look like mosses); Marchantia has rhizoids; gemmae detach and start new plants (asexual); stalks bear antheridia or archegonia; the sporophyte grows from the fertilized egg in the archegonium, and its capsule releases spores (23.2.17–18)
Hornworts	small horn-like sporophytes grow continuously from thin gametophytes (both photosynthesize); one chloroplast per sporophyte cell; moist, shaded places; cyanobacteria partners fix nitrogen; asexual fragmentation (23.2.19–20)
Mosses	largest group of nonvascular plants; moist places; shrivel and turn brown when dry, revive when it rains; peat, granite and true mosses; protonema → leafy shoots with rhizoids; sporophyte = foot + stalk + capsule, always attached (23.2.21–22)

Seedless vascular plants

Group	Key features
Lycophytes	roots (lower extensions of the stem); microphylls; strobili make spores by meiosis; homosporous or heterosporous; ground pines, spike mosses, quillworts (23.3.34–35)
Horsetails	Equisetum; wet, marshy places; reduced leaves in whorls at the nodes; strobili at stem tips; independent gametophytes; humidity-sensitive "spring-loaded" spores; silica makes stems tough (23.4.38–39)
Whisk ferns	Psilotum, Tmesipteris; no leaves (Tmesipteris: appendages like reduced megaphylls); horizontal rhizome; aboveground stems fork repeatedly; pumpkin-shaped sporangia on short side branches (23.4.40–41)
Ferns	mostly warm, moist tropics; fronds (megaphylls) divided into leaflets; sori on the undersides; dominant sporophyte; both generations independent (23.4.42–43)

Gymnosperm groups

Group	Key features
Conifers (most plentiful)	cones; pine needles with a thick cuticle and recessed stomata; windblown pollen; the seed is the dispersal stage; monoecious (23.5.58–59)
Cycads	large, finely divided leaves at the top of the stem; very large cones; pollen and seed cones on separate plants; insect-pollinated; multiflagellated sperm; plentiful in the Mesozoic (dinosaurs ate the seeds) (23.5.70–71)
Ginkgoes	one species, Ginkgo biloba; dioecious; pollution-resistant; multiflagellated sperm (23.5.72–73)
Gnetophytes	3 genera; similar xylem; no archegonia; similar strobili; Gnetum (trees or vines with leathery paired leaves), Ephedra (ephedrine), Welwitschia (two straplike leaves) (23.5.74–76)

Flower parts 23.5.83–85

Part	What it is / does
Sepals (together: calyx)	protect the bud; usually green; may drop off or look like petals
Petals (together: corolla)	vary in size, shape and color; attract a particular pollinator
Stamen (male)	anther (sac that makes pollen) + filament (stalk holding up the anther)
Carpel (female)	stigma (sticky knob) + style (slender stalk) + ovary (encloses the ovules)
After fertilization	ovule → seed; ovary → fruit

Already a table on the slides

Monocots vs eudicots: 23.5.81 (examples on 23.5.80). Flower terms (complete, perfect, inflorescence, composite): 23.5.91. Characteristics of each plant group: 23.1.4.

Look-alikes

These	The difference
antheridia vs archegonia	male gametangia (make sperm) vs female gametangia (make eggs) (23.1.12)
gametophyte vs sporophyte	n, makes gametes by mitosis vs 2n, makes spores by meiosis (23.1.12)
gametangia vs sporangium	where gametes are made (on the gametophyte) vs where spores are made (on the sporophyte) (23.1.12)
microphyll vs megaphyll	one unbranched vascular strand (lycophytes) vs branched veins (ferns, seed plants) (23.3.34, 23.4.36–37)
homosporous vs heterosporous	one type of spore vs two (microspores and megaspores) (23.3.34)
microspore vs megaspore	becomes the male gametophyte (pollen grain) vs the female gametophyte (23.3.34, 23.5.55)
strobili vs sori	cone-shaped spore-making structures (lycophytes, horsetails, gnetophytes) vs clusters of sporangia under fern fronds (23.3.34, 23.4.38, 23.5.74, 23.4.42)
rhizoids vs roots	hair-like threads that anchor and absorb (bryophytes, fern gametophytes) vs true roots (vascular plants; in lycophytes, lower extensions of the stem) (23.2.17, 23.3.34)
protonema vs prothallus	first, thread-like stage of the moss gametophyte vs the fern gametophyte (23.2.30, 23.4.52)
gemmae vs spores	groups of cells that detach from a liverwort (asexual) vs cells made by meiosis (23.2.17)
gymnosperm vs angiosperm	ovules not fully enclosed at pollination ("naked" seeds) vs ovules enclosed in an ovary that becomes a fruit (23.5.56–57)
monoecious vs dioecious	one plant makes both pollen and seed cones (conifers) vs separate plants (ginkgo; cycads' cones are on separate plants) (23.5.58, 23.5.70, 23.5.72)
pollination vs fertilization	pollen carried to the female part vs sperm uniting with the egg (23.5.55, 23.5.87–88)
calyx vs corolla	all the sepals vs all the petals (23.5.83)
stamen vs carpel	male (anther + filament) vs female (stigma + style + ovary) (23.5.84)
complete vs perfect	has all four flower parts vs has both stamens and carpels (23.5.91)
inflorescence vs composite	a cluster of flowers vs what looks like one flower but is many tiny ones (23.5.91)
zygote vs endosperm	sperm + egg, 2n vs sperm + 2 polar nuclei, 3n (23.5.88)
tube cell vs generative cell	grows the pollen tube vs divides into the 2 sperm (23.5.87, 23.5.89)
multiflagellated vs nonflagellated sperm	cycads and ginkgo (swim from the burst pollen tube) vs pine and flowering plants (23.5.70, 23.5.72, 23.5.68, 23.5.89)

Sequences

Plant evolution, in order 23.1.4, 23.1.6, 23.2.16, 23.3.32, 23.3.34, 23.4.36, 23.5.54–55, 23.5.70, 23.5.78

freshwater green algae (charophytes protect the zygote) → bryophytes (first on land; no vascular tissue; dominant gametophyte) → first vascular plants such as Cooksonia (branched: more sporangia) and lycophytes (vascular tissue, roots, microphylls; dominant sporophyte) → pteridophytes (megaphylls; dominant from the late Devonian through the Carboniferous) → gymnosperms (seeds and pollen; no external water for fertilization; cycads plentiful in the Mesozoic) → angiosperms (flowers, fruit, double fertilization; Triassic fossil pollen; Amborella is the closest living relative of the first ones)

Alternation of generations 23.1.12–13

sporophyte (2n) → meiosis in the sporangium → spores (n) → mitosis → gametophyte (n) → mitosis in the gametangia (antheridia: sperm; archegonia: eggs) → gametes (n) → fertilization → zygote (2n) → mitosis → sporophyte

How a megaphyll evolved 23.4.37

branched stem system → one branch dominates → side branches flatten into one plane → tissue fills the spaces between them → megaphyll

Life cycles, step by step

Printed in full once each: moss 23.2.30, fern 23.4.52, pine 23.5.68, flowering plant 23.5.89; compared side by side in the Life cycles table.

n / 2n / 3n and where meiosis happens

n / 2n / 3n 23.1.12–13, 23.2.17, 23.5.56, 23.5.86, 23.5.88–89

	Structures
n (haploid)	spores (microspores, megaspores); every gametophyte: moss leafy shoots and protonema, fern prothallus, pollen grain (male), embryo sac (female); antheridia and archegonia; eggs and sperm; the polar nuclei (each n); tube cell, generative cell
2n (diploid)	zygote; every sporophyte: moss foot + stalk + capsule, the fern plant, the pine tree, the flowering plant; sporangium; embryo; the ovary (sporophyte tissue) and the fruit it becomes
3n (triploid)	endosperm (sperm + 2 polar nuclei)
Meiosis (2n → n)	makes spores, in the sporophyte's sporangia: moss capsule; fern sporangia in sori; lycophyte strobili; pine pollen sacs and ovules; the flowering plant's pollen sacs (in anthers) and ovules
Mitosis	makes the gametes (in the gametangia); grows a spore into a gametophyte; grows the zygote into a sporophyte

Uses

Uses

Source	**Use (and why)**
Peat moss	improves the water capacity of soil; was a battlefield wound dressing; great commercial and ecological importance (23.2.21)
Ferns	edible ferns (food); Native Americans: astringent to stop bleeding in childbirth; florists, ornamentals (23.4.53)
Maidenhair fern	cold medicine (23.4.53)
Tropical tree ferns	building wood, because it resists decay, especially by termites (23.4.53)
Pines	parks and gardens (beauty, smell); needles and bark rich in vitamins A and C → teas for colds and coughs; pine nuts (large seeds) in cooking; pine oil scents products (23.5.69)
Ginkgo	pollution-resistant street and park trees; seeds a delicacy in Asia; extracts to improve blood circulation (23.5.72)
Ephedra	ephedrine, a medicine with serious side effects (23.5.74)
Angiosperms	all fruits, vegetables, nuts, herbs and grains (staples of the human diet); clothing, food, medicines (23.5.77)

Clue index: description → answer

Organisms and groups

closest green-algal relatives of land plants; they protect the zygote → **charophytes (Chara, Coleochaete)** 23.1.5
green alga with an unprotected embryo → **Oedogonium** 23.1.7
first plants to colonize land; no vascular tissue; low-lying → **bryophytes** 23.2.16
flat, ribbon-like body (thallus); gemmae; umbrella-shaped stalks with archegonia → **liverworts (Marchantia)** 23.2.17–18
horn-like sporophytes that keep growing; one chloroplast per cell; cyanobacteria partners → **hornworts (Anthoceros)** 23.2.19–20
looks dead and brown when dry, turns green after rain; largest nonvascular group → **mosses** 23.2.21
improves soil's water capacity; old battlefield wound dressing → **peat moss** 23.2.21
early fossil, a few cm tall, branched, sporangia at the branch tips → **Cooksonia** 23.3.32–33
leaves with one vascular strand; ground pines, spike mosses, quillworts → **lycophytes** 23.3.34
dominant from the late Devonian through the Carboniferous; megaphylls; no seeds → **pteridophytes** 23.4.36
silica in the stems; whorls of tiny leaves at the nodes; "spring-loaded" spores → **horsetails (Equisetum)** 23.4.38–39
no leaves; stems fork repeatedly; pumpkin-shaped sporangia → **whisk ferns (Psilotum)** 23.4.40–41
fronds; sori on the undersides; both generations independent → **ferns** 23.4.42–43
cold medicine → **maidenhair fern** 23.4.53
termite-resistant building wood → **tropical tree ferns** 23.4.53
"naked" seeds, not enclosed by a fruit → **gymnosperms** 23.5.56–57
needles with a thick cuticle and recessed stomata; one plant makes both cone types → **conifers (pines)** 23.5.58
vitamin A and C teas; pine nuts; pine oil → **pines** 23.5.69
insect-pollinated gymnosperm; huge cones on separate plants; dinosaurs ate its seeds → **cycads** 23.5.70
sperm with many flagella swim out of a burst pollen tube → **cycads and ginkgo** 23.5.70, 23.5.72
one surviving species; separate seed and pollen trees; survives city pollution → **Ginkgo biloba** 23.5.72
no archegonia; three genera → **gnetophytes** 23.5.74
source of ephedrine → **Ephedra (gnetophyte)** 23.5.74–75
only two huge straplike leaves that fray; southwestern Africa → **Welwitschia (gnetophyte)** 23.5.74, 23.5.76
trees or climbing vines with broad, leathery leaves in pairs → **Gnetum (gnetophyte)** 23.5.74
ovules always enclosed in diploid tissue; flowers and fruit → **angiosperms** 23.5.56, 23.5.77
living plant most closely related to the first flowering plants; New Caledonia → **Amborella** 23.5.78–79
long the earliest known angiosperm fossil, with fruit and flowers → **Archaefructus** 23.5.78
corn, tulips, pineapples, sugarcane → **monocots** 23.5.80
cacti, strawberries, dandelions, poplars, beans → **eudicots** 23.5.80

Structures

waxy, waterproof covering that prevents drying out → **cuticle** 23.1.9
small openings, mostly under leaves, for gas exchange → **stomata** 23.1.9
cells that carry water up against gravity → **tracheids** 23.1.9
where sperm / eggs are made on a gametophyte → **antheridia / archegonia (gametangia)** 23.1.12
where spores are made by meiosis → **sporangium** 23.1.12
hair-like threads that anchor and absorb, not true roots → **rhizoids** 23.2.17
groups of cells that break off a liverwort and grow into a new plant → **gemmae** 23.2.17
thread-like first stage of a moss gametophyte → **protonema** 23.2.30
the three parts of a moss sporophyte → **foot, stalk, capsule** 23.2.22
lid of a moss capsule; teeth that close it except in dry weather → **operculum** 23.2.30
strengthens the walls of xylem's conducting cells → **lignin** 23.3.31
carries water and minerals up / sucrose throughout the plant → **xylem / phloem** 23.3.31
small leaf with a single, unbranched vein → **microphyll** 23.3.34, 23.4.37
broad leaf with branched veins; collects more solar energy → **megaphyll** 23.4.36–37
cone-shaped structure that makes spores → **strobilus (plural strobili)** 23.3.34, 23.4.38
large fern leaf divided into leaflets → **frond** 23.4.42
clusters of sporangia under a fern leaf; the flap covering them → **sori; indusium** 23.4.42, 23.4.52
fern gametophyte with archegonia at the notch, antheridia at the tip → **prothallus** 23.4.52
horizontal stem that gives rise to fronds or aboveground stems → **rhizome** 23.4.40, 23.4.52
embryo + stored food in a protective coat → **seed** 23.5.54
male gametophyte of a seed plant → **pollen grain** 23.5.55
holds the female gametophyte; becomes the seed → **ovule** 23.5.55, 23.5.84
seed leaves that nourish the embryo → **cotyledons** 23.5.80
all the sepals / all the petals → **calyx / corolla** 23.5.83
slender stalk holding up the anther / sac that makes pollen → **filament / anther** 23.5.84
sticky knob / slender stalk / base enclosing ovules → **stigma / style / ovary** 23.5.84
female gametophyte of a flowering plant (usually 7 cells) → **embryo sac** 23.5.86, 23.5.89
two nuclei from opposite ends of the embryo sac → **polar nuclei** 23.5.86
small opening in the ovule where the sperm enter → **micropyle** 23.5.87
pollen cell that grows the tube / that divides into 2 sperm → **tube cell / generative cell** 23.5.87, 23.5.89
triploid food tissue in the seed → **endosperm** 23.5.88

Processes and terms

life switches between a many-celled n plant and a many-celled 2n plant → **alternation of generations** 23.1.12
the generation you notice (bigger, longer-lived) → **dominant generation** 23.1.14
why land plants have a diploid generation → **it hides the effect of a single deleterious allele (UV damage)** 23.1.11
makes one type of spore / two types → **homosporous / heterosporous** 23.3.34
seed plants keep their spores instead of releasing them → **seed plants (they retain the spores)** 23.5.55
wind or an animal carries pollen to the female part → **pollination** 23.5.55, 23.5.87
fertilization without outside water → **seed plants (sperm travel in a pollen tube)** 23.5.55
arrested state that lets a seed survive harsh conditions → **dormancy** 23.5.54
one sperm + egg, the other + polar nuclei → **double fertilization** 23.5.88
flower with all four parts / missing one or more → **complete / incomplete** 23.5.91
flower with stamens and carpels / only one of them → **perfect / imperfect** 23.5.91
cluster of flowers / looks like one flower but is many tiny ones → **inflorescence / composite** 23.5.91

one plant makes both pollen and seed cones → **monoecious (conifers)** 23.5.58
separate pollen-making and seed-making plants → **dioecious (ginkgo)** 23.5.72
plain flower, not showy → **wind-pollinated** 23.5.90
colorful flower → **bird- or insect-pollinated** 23.5.90
white or cream, fragrant, opens at night → **pollinated by nocturnal mammals or insects** 23.5.90
moist, fleshy covering (apple) / dry covering (acorn) → **fruits (from the ovary)** 23.5.88

Glossary: terms no slide explains

Adaptive radiation: one ancestral species quickly evolving into many new species that fit different ways of life. 23.5.78
Aerial stem: a stem that grows above ground. 23.3.35
Apical tissue: a patch of cells at the tips of roots and shoots that keeps dividing; it makes the plant grow longer and produces its other tissues. 23.1.4
Astringent: something that makes body tissues tighten, which helps stop bleeding. 23.4.53
Axil: the angle where a leaf joins the stem. 23.5.79
Capsella (shepherd's purse): a small flowering plant. 23.1.7
Clade: a group made of one common ancestor and all of its descendants. 23.1.5
Cladogram: a branching diagram showing how groups are related; each fork is a common ancestor. 23.1.6
Coleochaete: another charophyte; it grows as a flat disc of cells (pictured on 23.1.8). 23.1.5
Deleterious allele: a harmful version of a gene. 23.1.11
Desiccation: drying out. 23.1.9
Devonian and Carboniferous: periods in Earth's history, about 419-359 and 359-299 million years ago; the Carboniferous had huge swamp forests that later became coal. 23.4.36
Extant: still living today (not extinct). 23.5.57
Fitness: how many offspring an organism leaves; higher fitness means more offspring. 23.4.36
Fragmentation: a piece of the plant breaks off and grows into a new plant. 23.2.19
Gemma cup: a small cup on the liverwort's surface that holds gemmae until rain splashes them out. 23.2.18
Herbaceous vs woody: herbaceous plants have soft, green, non-woody stems; woody plants make wood, so their stems also grow thicker each year. 23.5.81
Indusium: a thin flap of tissue covering a sorus (cluster of sporangia). 23.4.47
Leaflets: the small separate blades that make up a divided (compound) leaf, like the many small segments of a fern frond. 23.4.42, 23.4.47, 23.4.52
Megasporangium (plural megasporangia): a sporangium that makes megaspores; in seed plants it is part of the ovule. 23.5.62
Megasporophyll: a leaf-like stalk that carries ovules or seeds. 23.5.73
Mesozoic era: the "age of dinosaurs," about 252-66 million years ago. 23.5.70
Nitrogen fixation: turning nitrogen gas from the air into a form living things can use. Plants can't do it themselves; some bacteria, like the cyanobacteria living in hornworts, can. 23.2.19
Node: the point on a stem where leaves or branches attach. 23.4.38–39
Oedogonium: a green alga made of threads of cells. 23.1.7
Ovule: a small structure where the female gametophyte (with its egg) develops; after fertilization it becomes the seed. 23.1.4, 23.5.55–56, 23.5.62 …
Paleobotanists: scientists who study fossil plants. 23.5.78
Phylum (plural phyla): a large group of related organisms, one level below kingdom. 23.2.21
Plant systematists: scientists who classify plants based on how they are related. 23.1.3
Polar nuclei: two nuclei in the middle of the embryo sac (the female gametophyte); the second sperm fuses with them to make the endosperm. 23.5.88
Pollen grain: the tiny male gametophyte of a seed plant; it carries the sperm to the egg. 23.1.4, 23.5.55
Pollen pore: an opening in the pollen grain's wall. 23.5.81
Pollen sacs (microsporangia): the sacs where pollen is made. 23.5.62–63
Pollen tube: a tube that grows out of the pollen grain and carries the sperm to the egg. 23.5.70, 23.5.72
Rhizome: a stem that grows sideways underground. 23.3.35, 23.4.39–41 …
Scale (of a cone): one of the flaps that make up a cone (woody in seed cones). 23.5.62
Scanning electron micrograph: a highly magnified picture of a surface taken with an electron microscope; "falsely colored" means the color was added afterward. 23.1.10
Silica: the hard mineral that sand and glass are made of. 23.4.38
Sterile cells: cells that don't make eggs, sperm or spores. 23.2.18
Substrate: the surface something grows on, like soil or rock. 23.1.12
Sucrose: table sugar; the form in which plants move sugar around. 23.3.31
Surface-area-to-volume ratio: how much outside surface something has compared with its size. Thin, branching shapes have a lot of surface, so they lose water fast. 23.1.11
Taproot vs fibrous roots: a taproot is one thick main root growing straight down (like a carrot); a fibrous root system is many thin roots of similar size (like grass). 23.5.81
Triassic, Jurassic and Tertiary: periods in Earth's history: about 252-201, 201-145 and 66-2.6 million years ago. 23.5.78
True roots, stems and leaves: ones with vascular tissue (xylem and phloem) inside. Bryophytes only have look-alike parts without it. 23.2.16
Uninucleated: each cell has one nucleus. 23.1.8
Vascular bundles: strands of xylem and phloem running through the stem. 23.5.81
Vascular tissue: tubes that carry water (xylem) and food (phloem) through the plant. 23.1.6, 23.1.14, 23.2.16, 23.3.32 …
Venation: the pattern of veins in a leaf: parallel (side by side, like grass) or net (branching, like a maple leaf). 23.5.81
Whorl: three or more leaves or branches growing in a ring around the stem at the same point. 23.4.38
Xylem and phloem: the plant's transport tissues. Xylem carries water up from the roots (its cells are lignified: stiffened with a tough substance called lignin); phloem carries sugars and other nutrients. 23.1.4, 23.5.74

Term index

Adaptive radiation see Glossary
Aerial stem see Glossary
Alternation of generations 23.1.12, 23.1.4, 23.1.13, 23.2.19
Amborella (angiosperm): closest living relative of the first flowering plants; New Caledonia 23.5.78–79
Angiosperms 23.5.56, 23.5.77, 23.1.4, 23.1.6, 23.1.14–15 …
Anther 23.5.84–85, 23.5.87, 23.5.89
Antheridia 23.1.12, 23.2.16–18, 23.2.22, 23.2.30 …
Apical tissue see Glossary
Archaefructus (fossil): long the earliest known angiosperm; fruit and flowers 23.5.78
Archegonia 23.1.12, 23.2.16–18, 23.2.22, 23.2.30 …
Astringent see Glossary
Axil see Glossary
Bryophytes (nonvascular plants) 23.2.16, 23.1.4, 23.1.6, 23.2.21
Calyx 23.5.83
Capsella (flowering plant): protected, nourished embryo in a seed 23.1.7
Capsule (moss) 23.2.22, 23.2.30, 23.2.17
Carpel 23.5.84–87, 23.5.89, 23.5.91
Chara (charophyte) 23.1.5, 23.1.8
Charophytes 23.1.5–6, 23.1.8
Clade see Glossary
Cladogram see Glossary
Club mosses, ground pines (Lycopodium) (lycophytes) 23.3.34–35
Coleochaete (charophyte) 23.1.5, 23.1.8
Complete / incomplete flower 23.5.91, 23.5.68
Composite flower 23.5.91
Conifers 23.5.58, 23.5.57, 23.5.59, 23.5.68–69
Cooksonia (early vascular plant fossil): branched, sporangia at the tips 23.3.32–33
Corolla 23.5.83
Cotyledons 23.5.80–81
Cuticle 23.1.9, 23.1.4, 23.1.10, 23.5.58
Cycads 23.5.70, 23.5.57, 23.5.71–72
Deleterious allele see Glossary
Desiccation see Glossary
Devonian and Carboniferous see Glossary
Dioecious 23.5.72
Dominant generation 23.1.14, 23.1.4, 23.1.11, 23.2.16, 23.3.32, 23.4.36 …
Dormancy 23.5.54
Double fertilization 23.5.88, 23.1.4, 23.5.89
Duckweed, Eucalyptus (angiosperms): the tiniest and the tallest 23.5.77
Embryo 23.1.3, 23.1.7, 23.5.54, 23.1.4, 23.5.68, 23.5.80 …
Embryo sac 23.5.86, 23.5.89
Endosperm (3n) 23.5.88, 23.1.4, 23.5.89
Ephedra (gnetophyte): source of ephedrine 23.5.74–75
Equisetum (horsetail): silica in stems; spring-loaded spores 23.4.38–39
Eudicots 23.5.80–82
Extant see Glossary
Ferns 23.4.42, 23.1.6, 23.1.14–15, 23.4.36, 23.4.40 …
Filament 23.5.84, 23.1.11, 23.5.85
Fitness see Glossary
Flagellated sperm 23.2.16, 23.5.70, 23.1.4, 23.2.30, 23.4.52, 23.5.68 …
Fragmentation see Glossary
Fronds 23.4.42, 23.4.52
Fruit 23.5.84, 23.5.88, 23.1.4, 23.5.56–57, 23.5.77 …
Gametangia 23.1.12, 23.1.4
Gametophyte 23.1.12, 23.1.14, 23.1.4, 23.1.13, 23.1.15, 23.2.16 …
Gemma cup see Glossary

Gemmae 23.2.17–18
Generative cell 23.5.89
Ginkgo biloba (the only ginkgo): dioecious; pollution-resistant 23.5.72, 23.5.57, 23.5.73
Gnetophytes 23.5.74, 23.5.57, 23.5.75–76
Gnetum (gnetophyte): trees or vines, leathery paired leaves 23.5.74
Granite mosses, true mosses 23.2.21
Green algae 23.1.3, 23.1.5–8
Gymnosperms 23.5.56–57, 23.1.4, 23.1.6, 23.1.15, 23.5.70 …
Herbaceous vs woody see Glossary
Homosporous / heterosporous 23.3.34, 23.1.4, 23.5.55, 23.5.86
Hornworts (Anthoceros) 23.2.19, 23.1.4, 23.1.6, 23.2.20
Horsetails 23.4.38, 23.4.36, 23.4.39
Indusium 23.4.52
Inflorescence 23.5.91, 23.5.79
Leaflets see Glossary
Lignin 23.3.31, 23.1.4
Liverworts 23.2.17, 23.1.4, 23.1.6, 23.2.18
Lycophytes 23.3.34, 23.1.4, 23.1.6, 23.3.31–32, 23.3.35
Maidenhair fern (fern): cold medicine 23.4.53
Marchantia (liverwort): rhizoids, gemmae 23.2.17–18
Megaphylls 23.4.36–37, 23.1.4, 23.4.38, 23.4.40, 23.4.42
Megasporangium (plural megasporangia) see Glossary
Megaspores 23.3.34, 23.5.55, 23.5.63, 23.5.68, 23.5.86, 23.5.89
Megasporophyll see Glossary
Mesozoic era see Glossary
Microphylls 23.3.34, 23.4.37, 23.1.4, 23.3.35
Micropyle 23.5.87
Microspores 23.3.34, 23.5.55, 23.5.63, 23.5.68, 23.5.87, 23.5.89
Monocots 23.5.80–82
Monoecious 23.5.58
Mosses 23.2.21, 23.1.4, 23.1.6, 23.1.14–15, 23.2.17 …
Nitrogen fixation see Glossary
Node see Glossary
Oedogonium (green alga): unprotected embryo 23.1.7
Operculum 23.2.30
Ovary 23.5.84, 23.1.4, 23.5.56, 23.5.85–86, 23.5.88 …
Ovule 23.5.55, 23.5.84, 23.1.4, 23.5.56, 23.5.62, 23.5.68 …
Paleobotanists see Glossary
Peat moss 23.2.21, 23.4.40
Perfect / imperfect flower 23.5.91
Petals 23.5.83, 23.5.82, 23.5.85, 23.5.91
Phloem 23.3.31, 23.1.4
Phylum (plural phyla) see Glossary
Pines (conifers) 23.5.58, 23.5.68–69, 23.1.14, 23.3.34, 23.5.59 …
Plant systematists see Glossary
Polar nuclei 23.5.86, 23.5.88–89
Pollen cones, seed cones 23.5.68, 23.1.4, 23.5.58–61 …
Pollen grain 23.5.55, 23.5.87, 23.1.4, 23.5.58, 23.5.64, 23.5.68 …
Pollen pore see Glossary
Pollen sacs (microsporangia) see Glossary
Pollen tube 23.5.55, 23.5.87, 23.5.68, 23.5.70, 23.5.72, 23.5.85 …
Pollination 23.5.55, 23.5.87, 23.5.90, 23.5.56, 23.5.68, 23.5.70 …
Prothallus 23.4.52
Protonema 23.2.22, 23.2.30
Psilotum (whisk fern): no leaves 23.4.40–41
Pteridophytes 23.4.36, 23.1.4, 23.3.32
Quillworts (Isoetes) (lycophytes) 23.3.34
Rhizoids 23.2.17, 23.1.15, 23.2.18, 23.2.22, 23.4.52
Rhizome 23.4.40, 23.3.35, 23.4.39, 23.4.41, 23.4.52
Scale (of a cone) see Glossary
Scanning electron micrograph see Glossary
Seed 23.5.54, 23.1.4, 23.1.6–7, 23.1.15, 23.5.55 …
Seed plants 23.5.54–56, 23.5.78
Seedless vascular plants 23.3.32, 23.1.4, 23.1.6, 23.4.36, 23.4.38
Sepals 23.5.83, 23.5.85, 23.5.91
Silica see Glossary
Sori 23.4.42, 23.4.52
Spike mosses (Selaginella) (lycophytes) 23.3.34–35
Sporangium 23.1.12, 23.1.4, 23.1.13, 23.2.22, 23.2.30, 23.3.32 …
Spores 23.1.12, 23.1.4, 23.1.13, 23.1.15, 23.2.17, 23.2.22 …
Sporophyte 23.1.12, 23.1.14, 23.1.4, 23.1.13, 23.1.15, 23.2.17 …
Spruce, juniper (conifers) 23.5.59
Stamen 23.5.84–85, 23.5.89, 23.5.91
Sterile cells see Glossary
Stigma 23.5.84–85, 23.5.87
Stomata 23.1.9–10, 23.5.58
Strobili 23.3.34–35, 23.4.38–39, 23.5.74
Style 23.5.84–85, 23.5.89
Substrate see Glossary
Sucrose see Glossary
Surface-area-to-volume ratio see Glossary
Taproot vs fibrous roots see Glossary
Thallus, thallose liverworts 23.2.17–18, 23.4.52
Tmesipteris (whisk fern): appendages like reduced megaphylls 23.4.40
Tracheids 23.1.9
Tracheophytes (vascular plants) 23.3.31, 23.1.4, 23.2.16, 23.2.21, 23.3.32–33 …
Tree ferns (ferns): termite-resistant wood 23.4.53, 23.4.44
Triassic, Jurassic and Tertiary see Glossary
True roots, stems and leaves see Glossary
Tube cell 23.5.87, 23.5.89
Uninucleated see Glossary
UV radiation 23.1.11, 23.1.9
Vascular bundles see Glossary
Vascular tissue 23.3.31, 23.1.4, 23.2.16, 23.3.32, 23.3.34, 23.4.36 …
Venation see Glossary
Welwitschia (gnetophyte): two straplike leaves; SW Africa 23.5.74, 23.5.76
Whisk ferns 23.4.40, 23.4.36, 23.4.41
Whorl see Glossary
Xylem 23.3.31, 23.1.4, 23.5.74
Xylem and phloem see Glossary
Zygote 23.1.13, 23.5.88, 23.1.3, 23.1.5, 23.2.25, 23.2.30 …